

Nitrate reduction across soils transitioning from coastal forest to wetland are hotspots for denitrification

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ABSTRACT

Sea level rise drives spatial migration of coastal ecosystems and can lead to the accelerated replacement of coastal forests with tidal wetlands. During this transition, changes in inundation frequency and saltwater intrusion dramatically alter soil biogeochemistry and redox conditions. Soil biogeochemical cycles in steady-state upland and wetland ecosystems are well studied, but pathways and rates in rapidly changing ecosystems are largely unconstrained. We characterized reduction of reactive nitrogen (N) via denitrification and dissimilatory nitrate reduction to ammonia (DNRA) at four sites where coastal forest is undergoing ecosystem state change and becoming wetland throughout the Chesapeake Bay. Sites were chosen to vary in soil characteristics, vegetation, and salinity regime. Soils were incubated with ^{15}N -labeled nitrate to estimate potential rates of denitrification and DNRA. Denitrification rates ranged from 0.011 to 26 $\text{nmol N g}^{-1} \text{hr}^{-1}$ and DNRA rates ranged from -0.30 to 5.6 $\text{nmol N g}^{-1} \text{hr}^{-1}$. DNRA rates were low with no clear pattern across the sites. Denitrification rates were higher in transition zone soils than in upland soils at three of the four sites ($p < 0.05$). Rates of denitrification in wetlands were lower than that observed in transition soils when hydrogen sulfide (H_2S) concentrations were high, favoring DNRA, but not where H_2S was low. Rates of denitrification across sites and zones were driven by soil moisture and related to organic matter content. Our data suggest that transition zone soils act as hotspots for N removal due to intermittent inundation and oscillating redox conditions. The N cycling characteristics of soils changing from forest to wetland are unique enough from steady-state upland and wetland ecosystems to address as a distinct ecotone in forecasts of future ecosystem function and N exchange at the coastal terrestrial-aquatic interface.

1. Introduction

Rising sea level is shifting terrestrial-aquatic interfaces landward (Ghanbari et al., 2019; Crowell, et al., 2010), submerging wetlands (Craft et al., 2009; Kirwan and Megonigal, 2013; Osland et al., 2022), and converting coastal upland forests to tidal wetlands (Kirwan and Gedan, 2019; Schieder and Kirwan, 2019). Changes in plant community composition are the most visible evidence of the landward movement of tidal wetlands, but driving plant responses, and more difficult to observe, are the altered inundation regimes and microbial responses that cause changes in soil redox status, cycling of nutrients, carbon, and metals (McNicol and Silver, 2014; Luo et al., 2019; Sengupta et al., 2021), and greenhouse gas emissions (Xu et al., 2021). Despite being critical to understanding coastal ecosystem responses to climate change, altered inundation depth and duration and their effects on exchanges of nutrients and carbon remain poorly characterized in ecosystems

undergoing rapid state change at the terrestrial-aquatic interface.

Soil nitrogen (N) cycles are distinctly different in upland forests, wetland forests, and herbaceous marshes at the coastal terrestrial-aquatic interface, which in turn regulate N availability, speciation, mobility, and N exchange with surface waters or the atmosphere (Seitzinger et al., 2006 and citations therein). N is often a limiting nutrient in coastal environments, a major contributor to coastal eutrophication (Howarth, 1988; Howarth and Marino, 2006), and a source of the powerful greenhouse gas N_2O . Primary production in forests (Sardar et al., 2023) and wetlands (Tobias and Neubauer, 2019) is N-limited (Sardar et al., 2023) and is sensitive to shifts in N cycle pathways (Purvaja et al., 2008). Changes to N cycling resulting from sea level rise at the terrestrial-aquatic interface remain unclear, but will likely be more complex than responses observed in forests subjected to freshwater flooding (Shrestha et al., 2014; Rennenberg et al., 2009) because of the increase in supply of redox-sensitive elements such as allochthonous

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nitrate (NO_3^-) and sulfate (SO_4^{2-}) that interact directly and indirectly with the N cycle and can be abundant in seawater. In addition, flooding also delivers allochthonous organic matter and O_2 , which can also influence N cycling at the terrestrial-aquatic interface.

Denitrification is a particularly important process that mitigates excess reactive N in the environment, controls primary production (Seitzinger et al., 2006), and regulates N_2O emissions. Denitrification is the stepwise reduction of reactive N species, such as NO_3^- , to inert dinitrogen (N_2) gas that is subsequently exported from the system. It is an anaerobic process that requires N in the oxidized form and, therefore, is often a dominant pathway at oxic-anoxic interfaces where nitrification and denitrification may be coupled (Jensen et al., 1994; Brune et al., 2000). In contrast, dissimilatory nitrate reduction to ammonium (DNRA) is a NO_3^- reduction pathway that retains reactive N in soils by converting it to ammonium (NH_4^+) (Giblin et al., 2013). DNRA has been shown to outcompete denitrification under high hydrogen sulfide (H_2S) conditions in saline wetlands (Murphy et al., 2020; Li et al., 2022) provided that the ratio of organic carbon to NO_3^- is relatively high (Megonigal et al., 2004; Hardison et al., 2015). Denitrification and DNRA are both influenced by soil geochemical properties such as organic content, N concentrations, soil moisture, the presence of H_2S (Li

et al., 2022), and wet-dry cycles (Groffman and Tiedje, 1988). The balance between these N recycling versus removal processes determines N availability, speciation, and hydrologic export (Wilson et al., 2023) at the terrestrial-aquatic interface.

Few studies have assessed NO_3^- reduction pathways in settings where coastal forests and adjacent wetlands are undergoing rapid state change, and comparisons across studies in single coastal ecosystems are limited by the use of distinctly different methods (Seitzinger et al., 2006). It is unclear how N cycling pathways may shift in soils undergoing conversion from upland coastal forest to tidal wetland. These are gaps in our understanding of coastal ecosystem state change that are best studied as continuums in the context of increasing inundation and salinity due to sea level rise (Ward et al., 2020). To address these knowledge gaps, we aimed to: (1) quantify potential denitrification and DNRA rates using consistent methods in coastal vegetated ecosystems that differ in inundation regimes, (2) compare NO_3^- reduction pathways across transects as a space for time substitution, and (3) determine geochemical drivers of the observed rates and the balance between N removal and recycling processes. Soils were collected along four forest-to-wetland transects in the Chesapeake Bay. The sites were chosen to provide contrasting controls on soil biogeochemistry with respect to soil characteristics,

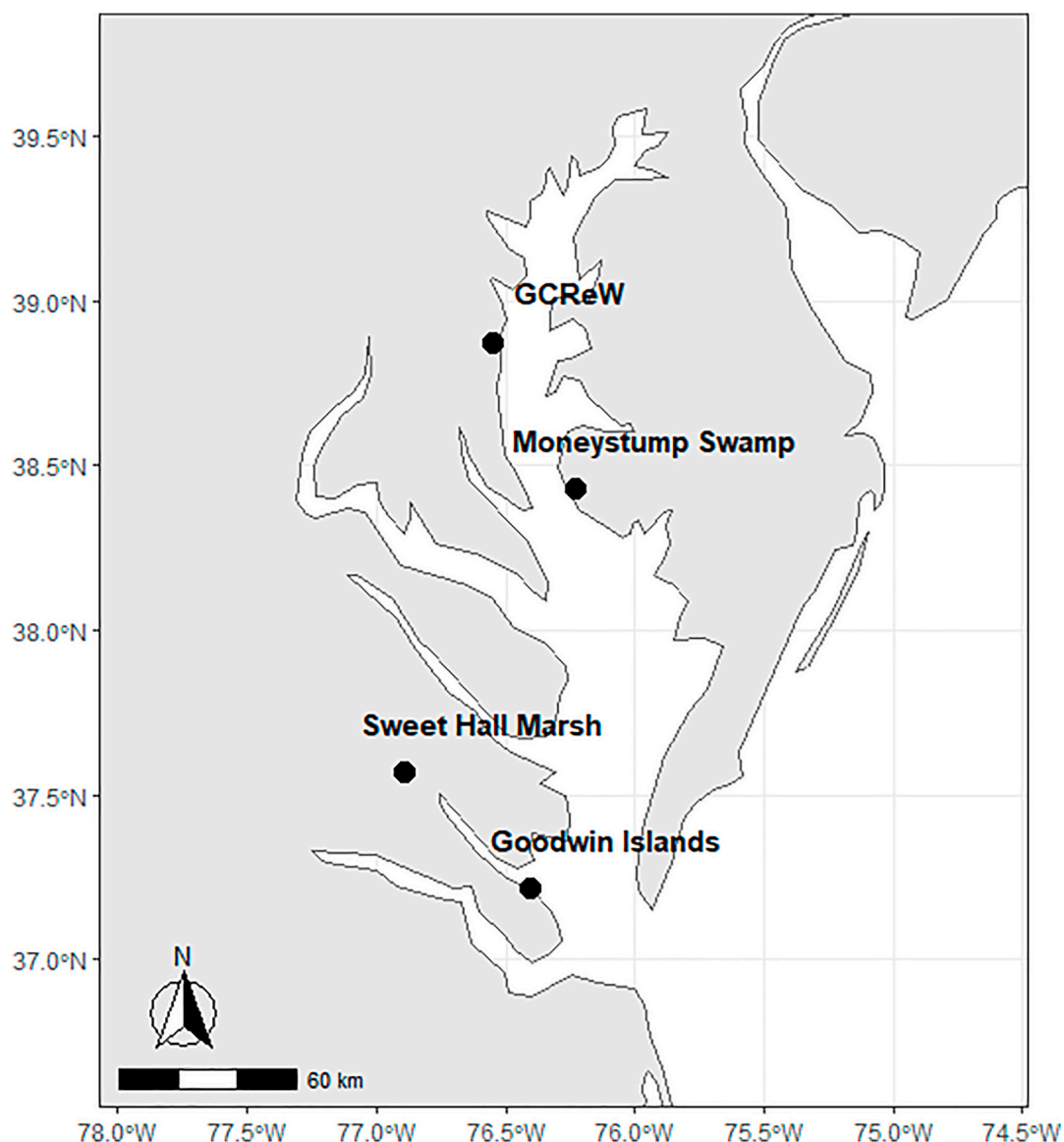


Fig. 1. Site map of sampling locations (GCRew, Moneystump Swamp, Sweet Hall Marsh, and the Goodwin Islands) throughout the Chesapeake Bay, USA.

vegetation, and salinity regimes. We hypothesized that soils changing from upland forest to wetland are supporting the highest rates of denitrification because of high-frequency oscillations between aerobic and anaerobic conditions that promote both nitrification (aerobic phase) and denitrification (anaerobic phase). Further, we hypothesized that DNRA is being favored by increasing salinity as a proxy for hydrogen sulfide (H_2S), resulting in lower denitrification:DNRA ratios.

2. Materials and methods

2.1. Study site locations

This study was conducted under the COMPASS-FME program (Coastal Observations, Mechanisms, and Predictions Across Systems and Scales—Field Measurements and Experiments), which has two focal regions, Chesapeake Bay and Western Lake Erie, USA (Ward et al., in review). We established four study sites in Chesapeake Bay (Fig. 1) selected primarily to represent a salinity gradient from freshwater (<0.5 psu) to polyhaline (~ 20 psu, Table S1), and secondarily to contrast geomorphic settings that vary in soil types and plant communities. At each site, we established one space for time transect that consists of three zones: an upland coastal forest (upland), a stressed forest transitioning to wetland (transition), and a tidal marsh (wetland). Sea level records, historical reconstructions, and visual observations of tree health establish that the transition zones at these sites were formerly upland forests in the past few decades (Ezer and Corlett, 2012; Schieder et al., 2018; Kirwan and Gedan, 2019; Walters et al., 2021; Smith and Kirwan, 2021). The persistence of the transition zone or further ecosystem state change to wetland will be largely controlled by inundation depth and duration. Water levels and salinity (Figs. S1 and S2) as well as soil moisture (Fig. S3) across the zones indicate the persistence of conditions unique within this ecotone between upland forest and wetland over two years.

Sweet Hall Marsh is a tidal wetland on the Pamunkey River. It is classified as a tidal freshwater marsh (salinity <0.5) but salinity is increasing with sea level rise, raising current surface water to up to 5 psu when river discharge is low (CBNERR-VA, 2022). The Sweet Hall site is unique in that the “upland” is a forested swamp; we will refer to it as the upland zone for comparison to other sites in this paper but note the differences in soil and redox conditions. The swamp and transition zone forests are dominated by red maple (*Acer rubrum*), and the wetland by arrow arum (*Peltandra virginica*) and big cordgrass (*Spartina cynosuroides*) among a high diversity of other herbaceous species.

The Moneystump Swamp site is located in the Blackwater Wildlife Refuge along Beaverdam Creek on the eastern shore of Maryland. Beaverdam Creek is mesohaline and ranges in salinity from ~ 5 to 10 psu. The upland zone is a loblolly pine (*Pinus taeda*) forest, and the marsh is dominated by a mixture of grasses including saltmeadow cordgrass (*Spartina patens*) and saltgrass (*Distichlis spicata*). The Global Change Research Wetland (GCREW) site is located in the Kirkpatrick Marsh in Edgewater, Maryland along the Rhode River. The Rhode River is also mesohaline and ranges in salinity from ~ 5 to 10 psu. The upland forest is a mixed deciduous tree community of sweet gum (*Liquidambar* spp.), red maple (*Acer rubrum*) and other species. The wetland area of GCREW used in this experiment is dominated by saltmeadow cordgrass (*Spartina patens*) and saltgrass (*Distichlis spicata*).

The Goodwin Islands site is located at the mouth of the York River Estuary, a tributary of the Chesapeake Bay. Surface water near the estuary mouth ranges from ~ 18 to 25 psu, the most saline study site included here (CBNERR-VA, 2022). The Goodwin Islands have maritime upland forest dominated by loblolly pine (*Pinus taeda*), the transition zone has a substantial phragmites (*Phragmites australis*) understory, and the wetland is dominated by saltmeadow cordgrass (*Spartina patens*) and saltgrass (*Distichlis spicata*). Soils at these sites range from well-drained upland soils to poorly drained organic-rich soils in the wetland. Among wetland zones, the Goodwin Islands and Sweet Hall sites have

organic-rich mineral soils while GCREW and the Moneystump Swamp have organic soils (Patel et al., in review).

2.2. Water level and soil moisture monitoring

Groundwater wells were installed in each zone to monitor groundwater levels, temperature, and conductivity across the sites. Groundwater wells were 5-cm-diameter slotted PVC pipes 1 m in depth. Each well was equipped with an Aquatroll600 (In Situ, USA) sonde that monitored water level, conductivity, temperature, dissolved oxygen (O_2), and pH at 15-min intervals. Sondes were calibrated according to manufacturer's instructions every three months. Soil moisture, conductivity, and temperature were also monitored in each zone with TEROS12 sensors (Meter, USA) logging at 15-min intervals. Sensors were installed at 10 cm and 30 cm with four replicates at 10 cm and two replicates at 30 cm in transition and upland zones. In wetland zones there were two replicates at 10 cm.

Surface water across sites, including salinity reported here, are monitored continuously with YSI EXO2 (Xylem, USA) water quality sondes. Sweet Hall Marsh and Goodwin Island site surface waters are monitored by the Chesapeake Bay National Estuarine Research Reserve water quality monitoring program (CBNERR-VA, 2022). GCREW site surface waters are monitored as part of the MARINE GEO program (Benson et al., 2022); represented as the Upper Chesapeake Bay site in the Rhode River. Lastly, Moneystump Swamp site surface water is monitored in the Beaverdam Creek as part of the COMPASS-FME synoptic site monitoring program (Pennington et al., in review; Ward et al., in review).

2.3. Porewater geochemistry

At three of the four sites (GCREW, Moneystump, and Goodwin Islands) vacuum lysimeters (Simpler Slim Tube Sampler, 1909L01, Soil Moisture, USA) were installed in each zone (upland, transition, wetland) for collecting porewater. Triplicate lysimeters were installed at 10, 20, and 45 cm (3 at each depth, 9 per zone). For this study, samples were collected from the 10 cm lysimeters only, except at the GCREW upland where porewater was collected from 20 cm due to a lack of water at the 10 cm depth. Porewater was collected within three days of soil coring and analyzed for salinity, SO_4^{2-} , H_2S , and dissolved inorganic nitrogen and phosphorus (NO_3^- , NO_2^- , NH_4^+ , PO_4^{3-}). Samples were filtered with 0.45 μm syringe filter (MilliporeSigma, USA). Samples analyzed for SO_4^{2-} and H_2S were preserved with 500 μL of 5 % Zinc Acetate and kept on ice until they could be refrigerated in the lab. Chloride and SO_4^{2-} concentrations were determined with an ion chromatograph (Dionex, Thermo Fisher Scientific, USA). Salinity (S, ppt) was calculated from chloride (Cl^- , ppm) concentrations in porewater samples using the Knudsen equation (Strickland and Parsons, 1972):

$$S = (1.807 * [\text{Cl}^-] + 0.026) / 1000 \quad (\text{Equation 1})$$

Porewater sulfide concentrations were determined colorimetrically using the Cline's method (Cline Joel, 1969). Porewater concentrations of NO_x ($\text{NO}_3^- + \text{NO}_2^-$), NH_4^+ and PO_4^{3-} were determined with a discrete autoanalyzer (SEAL, AQ300, USA) according to standard protocols.

2.4. Soil core collection & soil characteristics

At each site (GCREW, Moneystump, Goodwin Islands, and Sweet Hall Marsh) triplicate soil cores were collected in the upland, transition, and wetland zones. All sites were sampled during September of 2022 resulting in thirty-six total cores, nine per site. Core locations were determined at random by selecting a corner of the zone and generating three random distances from the plot corner to the core location. The top 10 cm of soil was collected with a 2.54 cm-diameter gouge auger after litter was removed from the soil surface. The cores were kept on ice until

returned to the lab where the full core was homogenized, removing root, leaves, and woody debris. A subsample of each homogenized core was frozen at -20°C for use in soil characterization and nutrient extractions. The remaining fresh soil was used immediately in soil slurry incubations to quantify soil NO_3^- reduction rates.

Soil bulk densities were sourced from soil characterization data collected along these transects by the EXCHANGE program (Pennington et al., 2023). Soil water content was determined by drying at 60°C for 48 h and reweighing. Organic matter content of dry soil was determined by loss on ignition (LOI) at 500°C for 4 h. Soil pH and conductivity were determined in slurries with a 1:5 soil-to-water ratio that were shaken for 1 h to equilibrate. Slurry pH was measured with an Orion™ 8102BN ROSS™ Combination pH Electrode probe (Thermo Fisher, USA). Soil conductivity was measured with a Multi240i meter equipped with a TetraCon 325/c conductivity probe (Xylem, USA).

Soil water extractions were conducted with 2 g soil, subsampled from thawed frozen soil, and 20 mL DI water, shaken for 30 min, then decanted and filtered (0.45 μm syringe filter, MilliporeSigma, USA). Extracts were analyzed for SO_4^{2-} and Cl^- concentrations using an ion chromatograph (Dionex, Thermofisher Scientific). Extractable nutrient concentrations were determined with potassium chloride (KCl) extractions. 2 M KCl was added to 4 g of soil, subsampled from thawed frozen soil, shaken for 1 h before decanting and the extract filtered with a 0.45 μm syringe filter (MilliporeSigma, USA). KCl-extracted samples were analyzed for inorganic nutrients including NO_x ($\text{NO}_3^- + \text{NO}_2^-$) with the EPA-127-D Rev A method (USEPA, 1993), NH_4^+ with the AGR-210-A method (USEPA, 1993), and PO_4^{3-} with the EPA-155-D method (USEPA, 1993) on a discrete autoanalyzer (SEAL AQ300). The NO_x and PO_4^{3-} methods were adapted for KCl extracts by preparing all standard, blank, and check standard concentrations in the extraction solution (KCl). Due to logistical constraints, extractions were performed on previously frozen soils, which may result in elevated nutrient concentrations compared to fresh soil extractions (Song et al., 2017); however, all soils were handled in the same fashion, so any artifacts should be consistent across samples. Extractable nutrient concentrations from fresh soils collected along the same transects at the GCRew, Money-stump, and Goodwin Island sites are available in Patel et al., (in review). In addition, soil pH, conductivity, gravimetric water content, total carbon, and total nitrogen in soils from the transects are available in Pennington et al. (2023).

2.5. $^{15}\text{NO}_3^-$ soil incubation experiments

Denitrification and DNRA rates were determined with the isotope tracing technique in soil slurry incubations, a method commonly applied to terrestrial soils (Semedo et al., 2018; Xi et al., 2016), wetland soils (Koop-Jakobsen and Giblin, 2010; Shan et al., 2016), and estuarine sediments (Fortin et al., 2021; Wilson et al., 2023). Although there are several other methods for measuring denitrification, isotope-labeled soil slurry incubations are applicable to the wide range of soil types and moisture conditions found across our coastal forest to wetland zones and are highly sensitive for quantifying denitrification (Groffman et al., 2006). It should be noted that these are potential rate measurements because samples were amended with NO_3^- and kept anaerobic, providing a measure of the capacity for N removal and recycling that is reflective of the *in situ* microbial capacity for denitrification and DNRA.

Incubations were conducted with duplicate subsamples of ~ 1 g composited soil from each homogenized core. Samples were pre-incubated in sealed helium-flushed (5 min), 12 mL exetainer vials (Labco, Lampeter, UK) for 24 h, then re-flushed and incubated for another 6–8 h at *in situ* soil temperature (19.6 – 25.5°C , Table S1) to allow for reduction of native $^{14}\text{NO}_3^-$. Samples were then flushed a second time with helium (5 min), amended with 100 nmoles $^{15}\text{NO}_3^-$ (99 atm%, Cambridge Isotope Laboratories, Inc., USA), and incubated at *in situ* soil temperature. Upland soils were incubated for 0, 2, 4, 6, and 10 h. Transition and Wetland soils were incubated for 0, 2, and 4 h. At each

time point, samples were spiked with 0.5 mL of 50 % zinc chloride and vortexed to terminate microbial activity.

To determine denitrification rates, $^{29}\text{N}_2$ and $^{30}\text{N}_2$ production over time was measured using a gas bench isotope ratio mass spectrometer (IRMS, Delta V Plus, Thermo Fisher Scientific, Waltham, MA, USA) as described by Song and Tobias (2011). Anammox rates were estimated from the $^{29}\text{N}_2$ production over time in soil incubations amended with $^{15}\text{NO}_3^-$ similar to Fortin et al. (2021); however, rates were very low and; therefore, the denitrification rates reported here are the $^{29+30}\text{N}_2$ produced over time to represent total reduction to N_2 (denitrification + anammox). Following gas analysis with IRMS, the same sample vials were used to determine DNRA rates. NH_4^+ was extracted from incubation samples using 5 mL of 2 M KCl, then diluted with Milli-Q (MilliporeSigma, USA) and poured into a new exetainer tube. Extracts were amended with 200 μL of hypobromite solution, which converts NH_4^+ to N_2 (Yin et al., 2014; Fortin et al., 2021), inverted, and incubated for at least 15 min before being analyzed with a membrane inlet mass spectrometer (MIMS, Balzers Prisma, Pfeiffer, Aßlar, Germany). The concentration of excess $^{29}\text{N}_2$ and $^{30}\text{N}_2$ produced by the addition of the hypobromite solution was calculated as described by (Risgaard-Petersen and Rysgaard, 1995) with the exception that a single air-equilibrated DI water standard, held at the same temperature as the samples, was used (Fortin et al., 2021). The concentrations of $^{29}\text{N}_2$ and $^{30}\text{N}_2$ were used to calculate the concentration of $^{15}\text{NH}_4^+$ present in each sample, and the production of $^{15}\text{NH}_4^+$ over time represented the DNRA rate. Rates were determined as the linear increase in the concentration of $^{29+30}\text{N}_2$ for denitrification or $^{15}\text{NH}_4^+$ for DNRA in both analytical replicate samples of each core measured over time.

For comparison to other studies, nitrate reduction rates were converted from a nmoles $\text{g}^{-1} \text{hr}^{-1}$ rate to areal rates in $\mu\text{moles m}^{-2} \text{d}^{-1}$ by converting nmoles to μmoles , hours to days, multiplying the 10 cm core increment by the bulk density of that soil (g cm^{-3}), and then converting from cm^2 to m^2 and are reported in Table S3. Data from other studies measuring denitrification and DNRA in soils were compiled from literature searches via Google Scholar and Web of Science for comparison to results in this study. Searches were conducted using key words including “soil”, “denitrification” and “DNRA” along with ecosystem identifying terms “forest” and “wetland”. The compiled rates were converted to relevant units when needed and are reported in Tables S7 and S8.

2.6. Statistical analyses

Data visualization and statistical analyses were conducted in R (R Core Team, 2021) using RStudio (RStudio Team, 2020). Results are presented as arithmetic mean $\pm$ standard error ($n = 3$ samples per site). A principle component analysis (PCA) of the observed denitrification rates, DNRA rates, and soil characteristics (soil pH, soil percent water, and soil organic content) was developed with the *prcomp* function in RStudio, and visualized with the *autoplot* function from the *ggfortify* package (Tang et al., 2016).

The nitrate reduction data was determined to be gamma distributed using the *fitdistrplus* package (Delignette-Muller and Dutang, 2015), log-likelihood, and AIC (Table S4) values. A generalized linear model (NO_3^- reduction $\sim$ site + zone + site:zone) was fit to the data using the *glm* function in R with a log link function and the gamma distribution. An ANOVA was used to assess statistically significant interactions between site and zones using the *car* package (Fox and Weisberg, 2019). Variation in soil NO_3^- reduction rates across sites and transects was assessed with estimated marginal means. The *emmeans* package in R (Lenth, n.d.) was used to conduct nonparametric comparisons of the *emmean* estimates via the *pairs* function, which employs a Tukey method to test the fixed effect of site (Sweet Hall Marsh, GCRew, Moneystump, and Goodwin Islands) and zone (upland, transition, wetland) on NO_3^- reduction rates via denitrification and DNRA ($\alpha = 0.05$, Tables S5 and S6). Pearson’s correlation (*cor* function) tests were used to assess correlations between soil NO_3^- reduction rates and soil characteristics.

3. Results

COMPASS-FME established the sites sampled in this study in 2022–2023. Here we report and summarize some of the first data that characterize the sites in a coherent study design (Ward et al., in review), then use the characterization data to interpret differences in N cycling pathways across the sites and infer mechanisms.

3.1. Soil characterization and porewater chemistry

Continuous monitoring at the study sites documented surface water salinities that follow the Chesapeake Bay salinity gradient and soil saturation patterns expected from the topographic gradients in our transect design. Salinity classes were oligohaline at Sweet Hall Marsh, mesohaline at GCREW and Moneystump Swamp, and polyhaline at the Goodwin Islands (Table S1). Soil porewater salinity followed patterns of variation in surface water salinity and flooding frequency and reflected elevation change across sites. Soil conductivity, measured in soil slurries, increased from upland to wetland, suggesting that estuarine water was more saline than groundwater even at the tidal freshwater site, Sweet Hall (Table 1). Depth and duration of saturation and inundation increased across the zones from upland to wetland (Fig. S1) as did groundwater salinity, following patterns of water depth (Fig. S2). Clear tidal cycles were observed at the Sweet Hall Marsh, a low-elevation site with a larger tidal range. The other wetland sites are high-elevation marshes that flood only on the highest tides. Average soil moisture increased from upland to wetland at all sites (Fig. S3) but was more variable in transition and wetland soils. The uplands at most sites had a volumetric water content of <0.5 (m³/m³), except at Sweet Hall Marsh where the “upland” zone is a forested swamp that regularly exhibits water contents >0.5.

A PCA of soil characteristics showed that our space-for-time substitution design captured hydrologic and biogeochemical differences that accompany sea level rise in Chesapeake Bay coastal ecosystems (Fig. 2). The PCA separated sites and zones by soil pH, water content, organic matter content (first axis), and NO₃⁻ reduction rates (second axis), explaining 79 % of the variation across samples. The PCA analysis reflected variation in soil characteristics such as water table depth and texture that did not always follow rank order across zones (Table 1). For example, gravimetric water content was higher in upland soils from Moneystump Swamp (45 %) and Sweet Hall (61 %) than from GCREW (11 %) and Goodwin Islands (27 %), and Sweet Hall and Goodwin Islands had higher water content in transition soils than wetland soils. These patterns contrast with the continuous data, reflecting spatial and temporal heterogeneity.

Upland soils tended to be more acidic than soils subject to occasional or frequent flooding at the same site, with upland pH ranging from 4.6 to 4.9 compared to 4.6–6.2 in transition soils and 5.3–6.3 in wetland soils

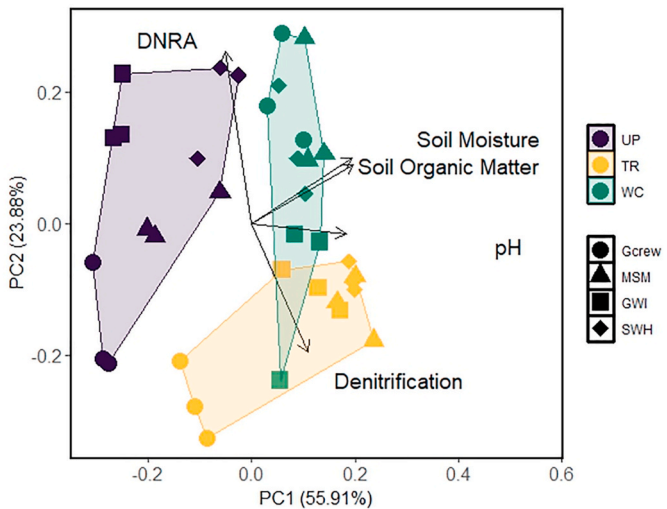


Fig. 2. PCA plot of soil characteristics (soil pH, percent organic content, and percent water) and N reduction rates (Denitrification and DNRA) measured in this study; indicates distinct zones across sampled terrestrial-aquatic interfaces (79 % variance explained by the first two axes). The first axis is explained by the soil characteristics and the second by nitrate reduction rates. Zones including Upland (UP), Transition (TR), and Wetland (WC) are denoted by color and sites GCREW, Goodwin Islands (GWI), Moneystump Swamp (MSM), and Sweet Hall Marsh (SWH) are denoted by point shape. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

(Table 1). Other features such as extractable nutrients did not show clear patterns. Extractable soil DIN concentrations ranged 21–140 μM with no clear upland to wetland pattern, and the extractable DIN pool is dominated by ammonium with very little NO₃⁻ or NO₂⁻ (Table 1). Similarly, porewater DIN concentrations ranged from 1.4 to 36 μM and were highly variable across transects and within zones with no cross-zone patterns.

Differences in salinity and flooding frequency across sites and zones affected sulfur cycling attributes due to SO₄²⁻ loading. Porewater SO₄²⁻ concentrations (Table S2) in the wetland zones increased with estuarine surface water salinities. H₂S concentrations in upland and transition zone porewater were uniformly low at all sites (<5.0 μM) but varied dramatically across wetland sites. Marshes at the mesohaline sites GCREW and Moneystump swamp had H₂S concentrations (920 μM and 170 μM, respectively) that were three orders of magnitude higher than Goodwin Islands (0.27 μM), the polyhaline site. These differences reflect the low soil mineral content of the mesohaline sites that would otherwise react rapidly with H₂S to form metal-sulfide precipitates. Low H₂S

Table 1

Site soil Characteristics; average values of three replicate cores used in the incubation experiments ± the standard error, n = 3. Soil pH and conductivity were determined in soil slurries. Water and organic content were determined by drying and loss on ignition and are reported as % of dry soil weight. Extractable dissolved inorganic nitrogen concentrations (μM) were determined with an autoanalyzer (see Methods).

Site	Zone	pH	Conductivity (μS cm ⁻¹)	Water Content (%)	Organic Content (%)	Extractable NO _x (nmoles/g)	Extractable NH ₄ ⁺ (nmoles/g)
GCREW	Upland	4.6 ± 0.04	23 ± 7	11 ± 1.1	21 ± 0.75	68 ± 15	2500 ± 240
	Transition	4.6 ± 0.02	49 ± 10	38 ± 0.71	46 ± 0.61	62 ± 5.7	2100 ± 250
	Wetland	5.3 ± 0.02	4300 ± 150	88 ± 0.58	96 ± 0.44	126 ± 86	3100 ± 830
Moneystump Swamp	Upland	4.7 ± 0.10	370 ± 52	45 ± 11	54 ± 13	146 ± 23	460 ± 190
	Transition	5.9 ± 0.11	3000 ± 160	86 ± 0.51	97 ± 0.15	43 ± 1.4	1500 ± 390
	Wetland	5.9 ± 0.17	5200 ± 93	90 ± 0.58	95 ± 2.2	64 ± 1.5	3900 ± 580
Goodwin Islands	Upland	4.7 ± 0.04	2200 ± 550	27 ± 1.5	34 ± 2.0	49 ± 0.25	1100 ± 290
	Transition	6.1 ± 0.10	4700 ± 110	69 ± 4.1	83 ± 5.2	177 ± 63	7000 ± 1100
	Wetland	6.3 ± 0.17	5500 ± 920	60 ± 9.7	69 ± 11	211 ± 25	6700 ± 1200
Sweet Hall Marsh	Upland	4.9 ± 0.07	250 ± 83	61 ± 5.1	71 ± 4.8	133 ± 71	2500 ± 290
	Transition	6.2 ± 0.05	1900 ± 190	88 ± 0.81	97 ± 0.22	44 ± 1.8	2100 ± 130
	Wetland	5.9 ± 0.08	2400 ± 180	77 ± 0.40	83 ± 0.49	39 ± 0.68	1100 ± 240

concentrations at Goodwin Islands may also reflect groundwater discharge of alternative electron acceptors such as ferric iron that suppress sulfate reduction. Other evidence of groundwater discharge at Goodwin Islands is the fact that surface water salinity in the adjacent estuary is 10 psu higher than at Moneystump, but soil conductivity in the wetland zones of the two sites were similar (Table 1). We did not collect porewater at the Sweet Hall site but assume that H_2S concentrations are uniformly low across the upland swamp, transition, and wetland zones as it is historically a tidal freshwater system.

3.2. Soil nitrate reduction rates

Potential soil denitrification rates ranged from 0.011 to 26 nmoles $\text{N}_2\text{-N wet g}^{-1} \text{h}^{-1}$ (Fig. 3, estimated marginal means presented in Table S5) and varied significantly by site and zone (zone, $p < 0.05$) with statistically significant interactions between site and zone ($X^2(6) = 70.8$, $p < 0.05$). At all sites, transition soils exhibited higher rates of denitrification than upland soils, though differences between zones were relatively small at Sweet Hall where the “upland” site is functionally a wetland. Wetland denitrification rates were highly variable and not consistently higher or lower than transition soils. Denitrification rates were correlated to soil moisture across the zones, but the direction of this relationship was opposite for wetland ($r = -0.92$) versus upland and transition ($r = 0.64$, Fig. 4) soils. A similar pattern was observed be-

tween denitrification and soil organic content with negative correlations in wetland soils ($r = -0.87$) versus positive correlations in upland and transition soils ($r = 0.63$, Fig. 5).

Potential DNRA rates ranged from -0.30 to $5.6 \text{ nmol N wet g}^{-1} \text{h}^{-1}$, with no clear pattern across sites or zones (Fig. 6). Although DNRA was observed across the zones, rates were not correlated to soil moisture (upland and transition soils, $r = -0.19$; wetland soils, $r = 0.08$), soil organic content (upland and transition, $r = -0.20$; wetland, $r = 0.03$), or other soil characteristics (Figs. S4–S6). In upland soils there was no clear pattern between soil moisture, organic content, and DNRA rates across sites, suggesting there may be other controls such as C and N content that determine the balance between N removal and recycling in coastal upland forests. The ratio of denitrification to DNRA ranged from 0.34 to 32 (Table 2). Upland soils exhibited the lowest total NO_3^- reduction rates, but dominance of denitrification versus DNRA was not consistent and varied by site. In transition zone soils, denitrification was the dominant pathway representing 71–97 % of the total NO_3^- reduction. In wetland soils, denitrification rates were higher than DNRA rates at Sweet Hall Marsh and Goodwin Islands but lower than DNRA rates at GCREW and Moneystump. Sites where DNRA rates exceeded denitrification were also sites exhibiting high porewater H_2S concentrations (GCREW and Moneystump). Anammox rates ranged from 0.0048 to $0.47 \text{ nmol g}^{-1} \text{h}^{-1}$ (Table S9), with no clear pattern across sites, and represented less than 9 % of total N_2 production in all samples.

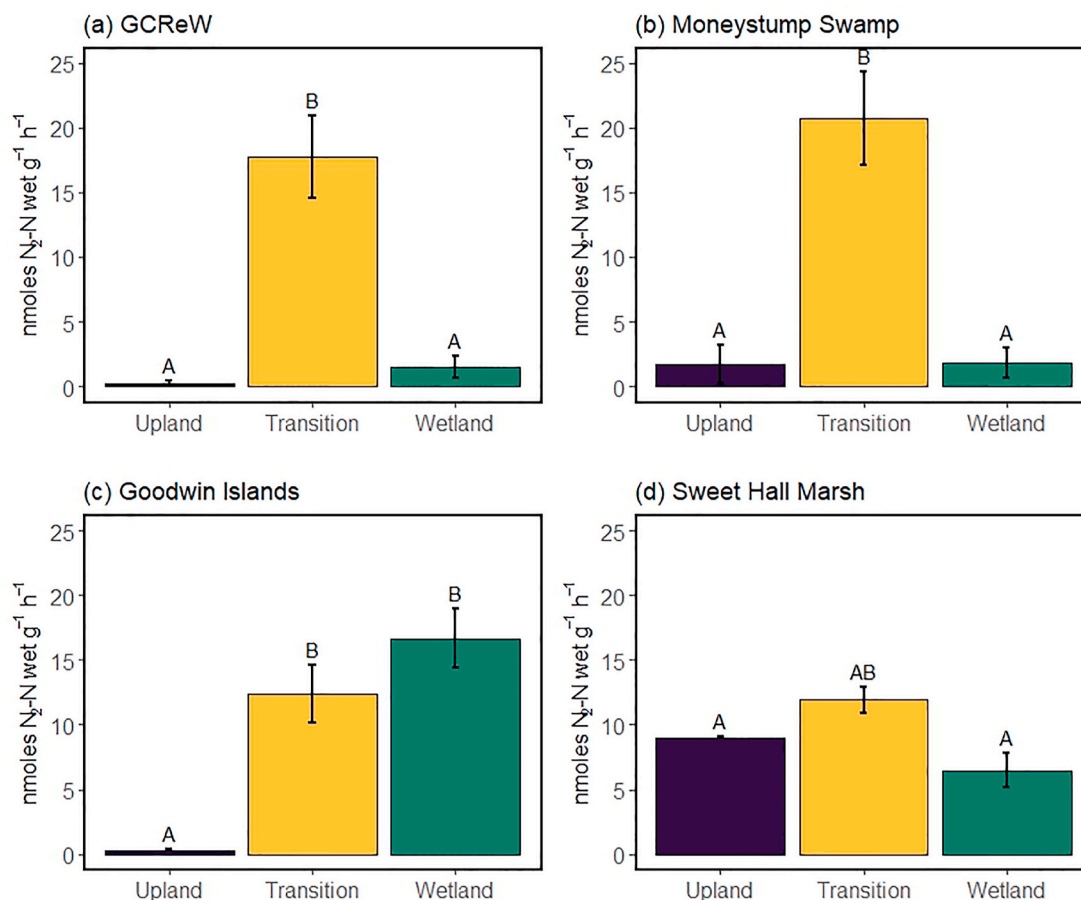


Fig. 3. Potential denitrification rates across zones at each site; (a) GCREW, (b) Moneystump Marsh, (c) Goodwin Island, and (d) Sweet Hall Marsh. Error bars indicate one standard error in each direction ($n = 3$). When zones do not share a letter, that indicates that there is a significant difference at $p \leq 0.05$ (Tukey HSD).

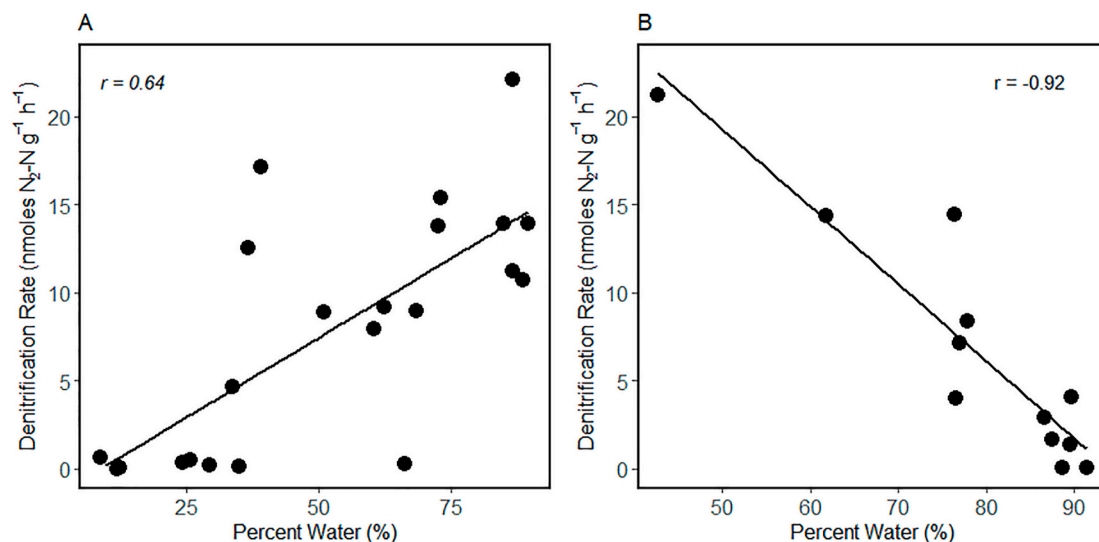


Fig. 4. Denitrification rates compared to soil moisture (percent water) in the upland and transition (A) and wetland (B) soils. Pearson correlation (r) between denitrification rates and soil moisture.

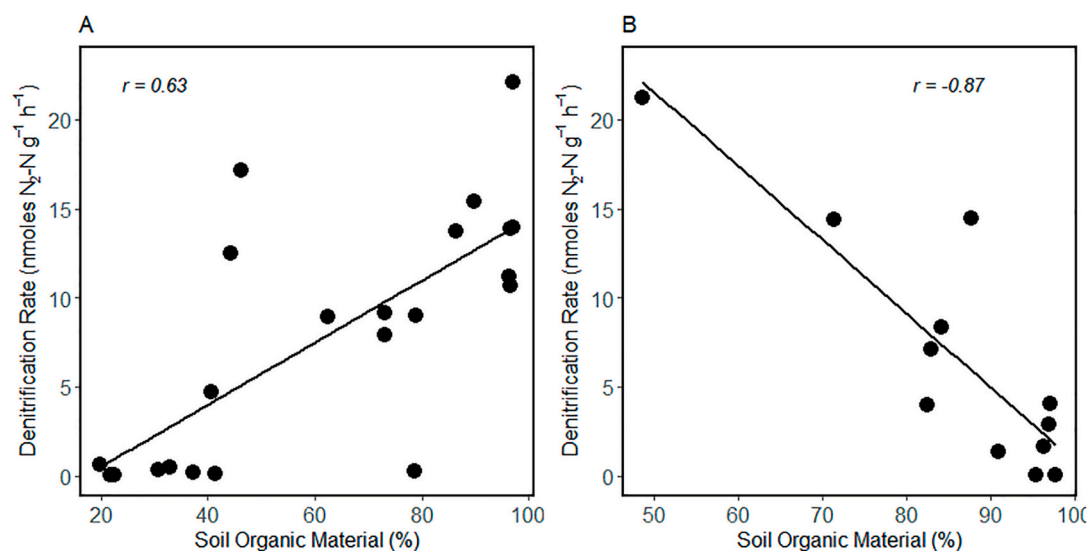


Fig. 5. A) Transition and Upland denitrification rates vs. soil organic material (determined by LOI). B) Wetland denitrification Rates vs. soil organic matter (LOI). Pearson correlation (r) between denitrification rates and soil organic matter.

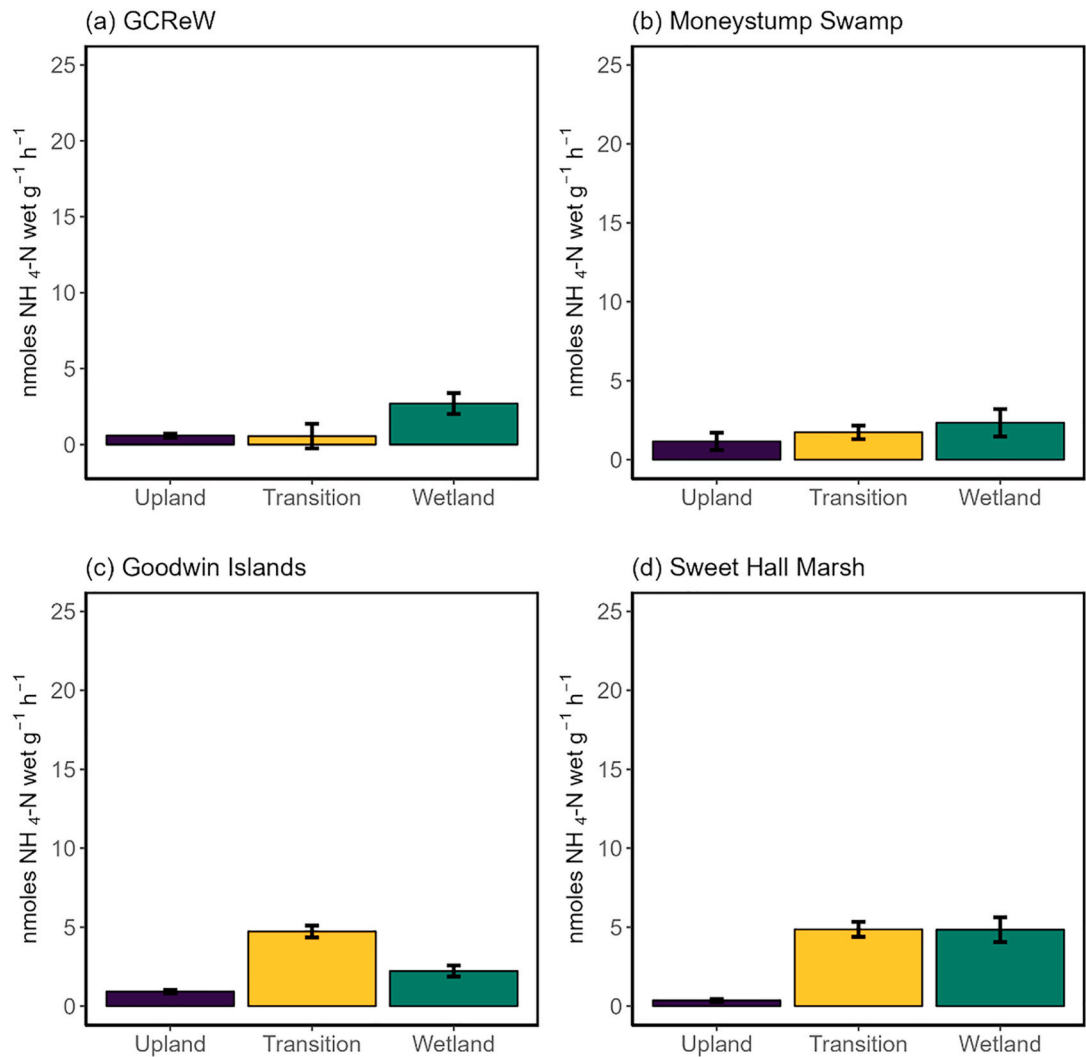


Fig. 6. Potential DNRA rates across zones at each site; (a) GCRew, (b) Moneystump Swamp, (c) Goodwin Islands, and (d) Sweet Hall Marsh. Error bars indicate one standard error in each direction. Rates were not significantly different from one another when comparing estimated marginal means with Tukey HSD.

Table 2
Denitrification and DNRA rates (arithmetic mean±standard error, n = 3) for each of the soils sampled, ratio of DNF to DNRA, and the % reduction by these processes to total nitrate reduction. Note here that when calculating the percent denitrification and DNRA the sum of these two pathways were assumed to be 100 % of total nitrate reduction, considering that the contribution of Annamox was always ≤9 % and <1 % in half of the estimates.

Site	Zone	Denitrification (nmol g ⁻¹ h ⁻¹)	DNRA (nmol g ⁻¹ h ⁻¹)	DNF:DNRA	% DNF	% DNRA
GCRew	Upland	0.22 ± 0.20	0.59 ± 0.12	0.4	27	73
	Transition	18 ± 3.2	0.56 ± 0.82	31	97	3
	Wetland	1.5 ± 0.82	2.7 ± 0.69	0.6	36	64
Moneystump Swamp	Upland	1.7 ± 1.5	1.2 ± 0.55	1.4	59	41
	Transition	21 ± 3.6	1.8 ± 0.43	11.8	92	8.0
	Wetland	1.8 ± 1.2	2.3 ± 0.93	0.8	45	55
Goodwin Islands	Upland	0.32 ± 0.09	0.93 ± 0.10	0.34	26	74
	Transition	12 ± 2.3	4.8 ± 0.38	2.6	72	28
	Wetland	17 ± 2.3	2.2 ± 0.35	7.5	88	12
Sweet Hall Marsh	Upland	9.0 ± 0.08	0.37 ± 0.09	25	96	4.0
	Transition	12 ± 1.0	4.9 ± 0.47	2.4	71	29
	Wetland	6.5 ± 1.3	3.0 ± 0.55	2.2	69	31

4. Discussion

4.1. Nitrate reduction rates across the terrestrial aquatic interface

The balance between soil denitrification and DNRA determines whether N will be removed or recycled at the coastal terrestrial-aquatic interface. Across the sites, the ratio of denitrification to DNRA ranged by

orders of magnitude, from 0.34 to 32, but with no clear pattern from upland to wetland. In soils changing from upland to wetland (i.e., transition zone soils), denitrification was the dominant NO₃⁻ reduction pathway, contributing over 70 % of total NO₃⁻ reduction at all sites (Table 2). This suggests that N removal may dominate during the conversion period when upland forests are receiving increased inputs of allochthonous N from estuarine surface water as inundation frequency

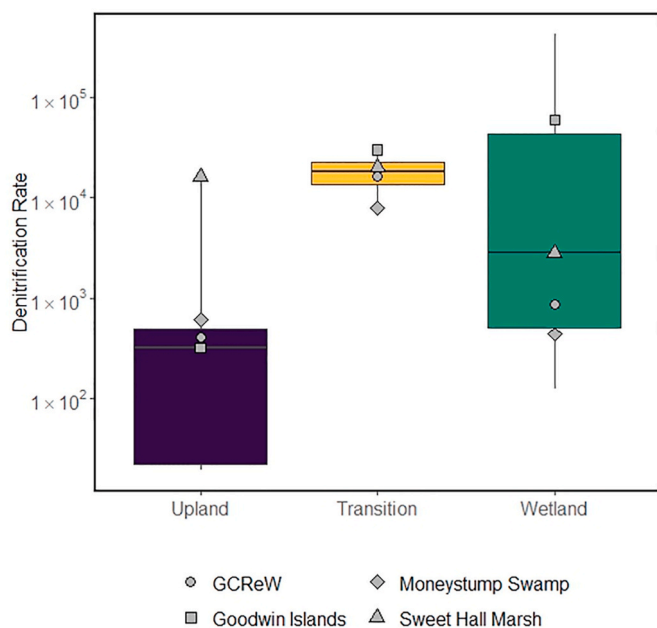


Fig. 7. Box and whisker of log denitrification rates compiled from the literature, see Table S7, and this study for uplands ($n = 15$), transitions ($n = 4$, this study only), and wetlands ($n = 19$). Grey points indicate rates measured in this study. It should be noted that floodplain forests and not well drained upland soils were excluded from the upland zone here due to their wetter soils, but the Sweet Hall site has been retained and is identified by the symbols.

and duration increase. However, our data also suggest that as transition zones become wetlands, the balance between these processes will change further depending on the presence of redox sensitive elements such as Fe in soils or changes in SO_4^{2-} loading.

An advantage of our study design is that it allows comparisons of N-cycling rates across upland, transition, and wetland zones using consistent methodologies, as opposed to most of the existing literature where the methods applied vary across ecosystem compartments. In comparison to the literature, soils in our upland zone exhibited rates in the top 25 % quartile of upland denitrification rates (Fig. 7, Table S7). The only exception was Sweet Hall, where rates were greater than all compiled upland rates, presumably because the “upland” zone of the Sweet Hall site is a forested swamp and, therefore, functionally a wetland. The upland zones of sites other than Sweet Hall are rarely inundated, have high redox potential, and are typically aerobic, so low rates of NO_3^- reduction are expected compared to transition and wetland soils (Parkin and Tiedje, 1984). Wetland denitrification rates were below the median of compiled wetland rates, except for those measured at the Goodwin Islands where rates exceeded the median (Fig. 7). Goodwin Islands denitrification rates in the wetland and transition zones were the same order of magnitude as rates reported by Ooi et al. (2022) in New England *Spartina* and *Phragmites* marshes, species present at high abundance at the Goodwin site. Compared across zones, denitrification rates were higher in soils that are actively shifting from forest to wetland (transition zones) than those in upland forest soils (Fig. 3). This pattern was consistent at all sites, from these data we can infer that transition zone soils develop unique N cycles and are hotspots for gaseous N export. Wetland soils also have the potential to be hotspots of NO_3^- reduction, as illustrated by the Goodwin Islands and Sweet Hall sites (noting that the “upland” site in the latter is functionally a wetland), where H_2S production is either low (Sweet Hall) or H_2S is scavenged by reaction with soil minerals. However, at sites where H_2S accumulates in porewater due to an absence of minerals (i.e., GCREW and Moneystump, see section 4.2), NO_3^- reduction was suppressed favoring DNRA.

Rates measured in this study reflect potential denitrification and

DNRA rather than *in situ* rates because the samples were incubated under anaerobic conditions and amended with NO_3^- , conditions that favor denitrification and DNRA. Under such conditions, rates are assumed to reflect the biomass of the active denitrifier and DNRA microbial communities, and thus the influence of site conditions on the capacity for N removal or recycling by these pathways in the days and weeks preceding the soil sample (Wilson et al., 2023; Marton et al., 2012). Although potential denitrification rates peaked in the transition zone, rates can change on relatively short time scales as C and NO_3^- substrate availability respond to soil saturation, O_2 concentration, and redox status. Potential rates in the present study may be underestimated because the amount of NO_3^- added to the incubations was quite low (100 nmoles), in an attempt to limit deviations from *in situ* conditions. That being noted, we estimate that our amendments increased the initial extractable soil NO_3^- , which ranged from 38 to 287 nmol g^{-1} soil, by 35–263 %. Based on the strong and consistent pattern of peak denitrification rates in transition zone soils to substantial NO_3^- amendments under anaerobic conditions, we conclude that variable hydrologic conditions and oscillating aerobic-anaerobic conditions in the transition zone likely favor relatively large denitrifier communities in areas of the coastal terrestrial-aquatic interface where upland forests are being converted to tidal wetland.

DNRA rates across zones were within the range of previously reported rates in forests and wetlands (Table S8) but varied widely and exhibited no clear pattern across the zones in our study. For example, at two sites potential DNRA rates were similar in magnitude in upland versus wetland soils, while transition zone soils exhibited both the maximum and minimum mean rates observed across the zones (Fig. 3). Compared to denitrification where soil water content and organic matter content variation are dominant rate-controlling factors, it seems the processes controlling DNRA appear more complex, other than suppression of denitrification in the presence of H_2S . As one goal of COMPASS is to support predictive numerical models of N cycling processes, further research is required to understand biogeochemical control of DNRA. Our data indicate that NO_3^- reduction (both to removal or recycling) is low in these coastal upland forests, denitrification is dominant in the transition zones, and DNRA will dominate in wetlands where conditions favor high porewater H_2S concentrations.

Other pathways for N reduction include annamox, fungal denitrification, and abiotic co-denitrification. Annamox has been shown to remove N at the terrestrial-aquatic interface at rates that account for <3 % of N_2 production in wetlands (Koop-Jakobsen and Giblin, 2009) and <15 % in forest soils (Xi et al., 2016). This aligns with measurements in our study where annamox represented 0.22–9.0 % of N removal. Based on low activities compared to denitrification and DNRA, annamox is a minor component of N reduction in these soils. We did not measure fungal denitrification (Bösch et al., 2023) or abiotic co-denitrification which have been shown to reduce N under acidic conditions or when NO_3^- concentrations are high (Phillips et al., 2017). It is difficult to parse abiotic from biotic N_2 production, but because the soil pH in this study was ≥ 4.6 abiotic co-denitrification was likely limited (Wilson et al., 2021).

4.2. Inundation frequency and redox sensitive elements control nitrate reduction

Our data suggest that control mechanisms for NO_3^- reduction are fundamentally shifting as coastal upland forests at the terrestrial-aquatic interface continue to undergo ecosystem state change, illustrated clearly by the opposite direction (sign) of the correlations between denitrification and soil water content in upland and transition zone forest soils versus wetland soils (Fig. 4). Denitrification in upland soils is likely limited by high O_2 availability which favors aerobic respiration in the facultatively anaerobic denitrifying bacteria that tend to dominate the process. Low rates in upland ecosystems are consistent with previous work on terrestrial (freely drained) soils where denitrification accounts

for only ~22 % of N removal (Seitzinger et al., 2006, Fig. 6). Dependence of denitrification on a limited O_2 supply is a potential mechanistic (cause and effect) explanation for the positive correlation to soil moisture observed in upland and transition zone soils (Fig. 4) because water is an effective barrier to O_2 diffusion. In addition, because denitrification is a heterotrophic process and potentially C limited, the correlation with soil organic matter may also reflect a mechanistic link to the electron donor supply; however, considering that organic content is correlated to soil moisture, we speculate that its relationship to denitrification is simply that of a covariate. The change in the sign of the correlation between denitrification and soil water content that flipped to negative in tidal wetland soils suggests NO_3^- limitation develops with longer and more frequent periods of soil saturation. NO_3^- limitation is more likely than C limitation in the absence of an external NO_3^- source.

Transition zone soils exhibited the highest denitrification rates in both the present study using consistent methods and the literature survey where methods varied across zones. This is explained by the presence of abundant and balanced substrates for denitrification compared to the wetland and upland extremes and, by inference, relatively large denitrifying microbial communities. Transition zone soils experience water level fluctuations that allow for periods of high O_2 concentrations that support nitrification followed by periods of low O_2 concentrations that favor denitrification. This aligns with mesocosm experiments showing that wet-dry cycles enhance soil denitrification rates (Tomasek et al., 2019; Baldwin and Mitchell, 2000; Corstanje and Reddy, 2004). Cycling between saturated and unsaturated conditions, as well as lateral connectivity across landscapes, have been proposed as mechanisms for mitigating N pollution based on the concept of optimum conditions for denitrification (Tomasek et al., 2019; Kaden et al., 2021). Our data support this concept and suggest that soils undergoing change from coastal forest to wetland will begin to export increasing amounts of N_2 , offsetting some fraction of the increased N inputs from increased inundation. The net effects of these changes on ecosystem N stocks are uncertain without quantifying change in allochthonous N inputs, but it suggests that forests becoming wetlands will, at least initially, remove increasing amounts of reactive N from external loading into the system.

Differences in the availability of redox-sensitive elements interacted with patterns of water level fluctuations to regulate N removal pathways. Wetlands with the highest soil water and organic contents – GCRew and Moneystump – were also sites that exhibited the highest porewater H_2S concentrations and lowest denitrification rates. In contrast, wetland soils from Goodwin Islands and Sweet Hall Marsh had low H_2S concentrations and relatively high denitrification rates. H_2S inhibits denitrification (Sørensen et al., 1980) and is a dominant control on NO_3^- reduction pathways in brackish marshes, causing a shift away from denitrification toward DNRA (Murphy et al., 2020). This mechanism is favored in highly organic soils such as those at GCRew and Moneystump, which are poor in iron minerals to which sulfide would otherwise rapidly and nearly irreversibly bind, forming FeS and Fe_2S (Sundelin and Eriksson, 2001). Some of our results were counterintuitive as the most saline and sulfate-rich site – Goodwin Islands – had lower porewater H_2S concentrations than our mesohaline sites. This may be explained by one or a combination of three mechanisms: (i) relatively mineral-rich soils at Goodwin Islands that rapidly remove H_2S via complexation, (ii) microbially available Fe(III) in soil minerals that supports microbial iron reduction and suppresses sulfate reduction and reducing available H_2S , and (iii) relatively limited SO_4^{2-} availability due to fresh groundwater discharge emerging in the wetland. We are not yet able to distinguish between these mechanisms but suspect they are all operating to some degree.

The mineral content of coastal wetland soils is strongly controlled by a variety of hydrogeomorphic factors such as suspended sediment supply, distance from tidal creeks, soil surface elevation, and wetland age, all of which are highly dynamic in coastal ecosystems as they respond to accelerated sea level rise. Our data indicate that upland ecosystems undergoing state change and becoming wetlands will continue to favor

the denitrification pathway over DNRA due to the initially high mineral content of upland soils. We predict that the potential for wetlands to switch to favoring DNRA will occur when the organic-rich and microbially active soil surface becomes separated from the legacy upland soil horizon due to accretion and burial, and in settings where surficial mineral inputs from inundation are limited by marsh elevation or distance from tidal creeks. Because marsh elevation models are capable of making robust predictions of soil mineral concentrations, numerical tools already exist to predict the time scale at which this transition will occur, supporting the forecast goals of COMPASS-FME. DNRA is typically favored under low NO_3^- and high organic carbon conditions as it transfers a higher number of electrons per mole of NO_3^- (Tiedje, 1988; Megoñigal et al., 2004). However, there were no clear relationships between the variables we measured and DNRA within or across transects.

4.3. Implications for biogeochemical cycling at the terrestrial-aquatic interface in the face of sea level rise

As upland ecosystems become wetlands, the organic content and presence of H_2S in soils will determine their capacity for N removal versus recycling. The introduction of saline water to previously fresh systems makes SO_4^{2-} available for sulfate reduction producing H_2S . As shown here, the accumulation of H_2S plays a critical role in the ratio of N removed to N recycled. Saltwater intrusion into tidal fresh wetlands has been linked to increased reactive N in porewaters, H_2S accumulation, and decreased plant productivity (Widney et al., 2019). The timeline over which these ecosystems undergo ecosystem state change varies with physical factors that determine inundation frequency such elevation gradient and the local rate of sea level rise (Kirwan and Megoñigal, 2013; Ward et al., 2020). The order-of-magnitude variation across sites in the ratio of denitrification to DNRA we observed suggests that timelines will also be heavily influenced by legacy effects such as soil organic content and mineralogy, and by hydrogeomorphic factors that interact with climate to govern salinity.

NO_3^- reduction yields nearly as much free energy as aerobic respiration, making N-cycling processes a dominant control on the cycling of carbon and organically bound nutrients and other elements. The role of NO_3^- reduction in regulating the redox status of wetland soils is typically limited by NO_3^- availability unless there is a source of NO_3^- loading such as agricultural runoff. Future work on the biogeochemical consequences of ecosystem state change in coastal zones should focus on the consequences of NO_3^- depletion for the cycling of alternative electron acceptors such as Mn(III,IV), Fe(III), and SO_4^{2-} . Ultimately, coastal ecosystems undergoing transition will become poised at a redox state set by carbon availability (i.e., electron donors) and the relative supply of electron acceptors as determined by factors such as soil characteristics, substrate inputs and outputs to the system, and internal reoxidation, all of which vary strongly in space and time. From a biogeochemical perspective, such variation does not separate cleanly into upland, transition, and wetland categories and is perhaps more profitably conceptualized as a continuum.

4.4. Conclusions

Controls on N cycling at the coastal terrestrial-aquatic interface will shift as ecosystems change state. Increasing periods of soil saturation and changes in plant communities are visible evidence of such changes. Less apparent are major shifts in biogeochemical cycles and the proportion of the soil profile supporting aerobic versus anaerobic processes. In the case of N cycling, our data suggest that the balance of soil N removal and recycling are shifting, but the shifts will not be monotonic with saturation and are likely to be highly variable depending on the geomorphic setting (Kirwan et al., 2023). From our data, we can infer that as soils are changing from upland to wetland, denitrification is increasing due to frequent oscillations between oxic and anoxic

conditions, a recognized characteristic of landscape hotspots of denitrification (Groffman et al., 2006). However, as these systems continue to further convert, becoming regularly inundated tidal wetlands, trends in denitrification are reversing and rates are decreasing. Our results suggest that transition zones provide unique niches for biogeochemical cycling and, therefore, function distinctly from upland and wetland ecosystem states.

CRediT authorship contribution statement

Stephanie J. Wilson: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **J. Patrick Megonigal:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Data availability Statement

Data leveraged in this manuscript, data analysis code, and associated files are available through the COMPASS GitHub repository via: https://github.com/wilsonsj100/TAI_NCycle. Continuous monitoring data included in this manuscript are available via ESS-DIVE: COMPASS-FME Synoptic Sites Level 1 Sensor Data v1-1. <https://doi.org/10.15485/2439400>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.soilbio.2025.109904>.

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